

BUSINESS CASE	
Proposed Project	AI Learning Environment (working name)
Date Produced	6 October 2026
Background	University students do most of their studying from course files they have already downloaded: lecture slides, textbook chapters and assignment PDFs saved from UR Courses into per-course folders on a laptop or iPad. Many students now also use AI assistants (ChatGPT, Claude, Gemini) and study tools (NotebookLM, Quizlet, Anki) alongside those files. These tools are disconnected from the files. To get help with a page, a student uploads the file or pastes the passage into a chat, explains which course it is from, and repeats that setup in every new session. The answer usually does not point back to the page it came from, so the student cannot easily check it. Practice material such as flashcards and quizzes is made separately, by hand, in another app. The project came out of a primary-research interview with Linda, a U of R student, who described this as the main friction in her study sessions: "it would be smoother if it could see the file I am studying." It was selected on 15 September 2026 from three ideas the team pitched in ENSE 400. The other two were uofrEats (campus food ordering and pickup) and AI-assisted drones for crop scouting and fire detection. The AI Learning Environment was chosen because: • It is backed by real evidence: the problem statement comes directly from a student interview rather than an assumption. • It needs no hardware or outside partners: the drones idea depends on hardware and flight regulations, and uofrEats depends on signing up campus restaurants. • The team has the problem too: both members study from PDFs every week, so they can test the product on their own coursework. The project is being built and self-funded by Aubin Chriss Izere and Chop Kur as their two-term Software Systems Engineering capstone (ENSE 400 in Fall 2026, ENSE 477 in Winter 2027) with Dr. Tim Maciag.
Business Need/ Opportunity	Business need • Repeated setup: students lose time and focus re-supplying the same files and course context at the start of every AI study session. • Unverifiable answers: explanations are not tied to the page being read, so students cannot quickly check them against the source. • Manual practice: flashcards and quizzes are made by hand in a separate app, and that effort is the main reason students stop using them. Opportunity No current tool combines three things: working from the course folders already on the student's device, explanations anchored to the exact page, and practice generated from the same passage. • NotebookLM needs every file uploaded and runs only in the cloud. • Adobe Acrobat AI Assistant works one document at a time and has no practice features.

	• Goodnotes AI is notebook-centred, iPad-first and weaker on desktop. • General assistants (ChatGPT, Claude, Gemini) need context set up by hand every session. • Anki and Quizlet have no explanations, and cards are made by hand. The starting market is U of R undergraduates who study from PDFs, with graduate and thesis students as a second group. Running the free tier on free cloud AI models means students can start at no cost, and the team pays nothing per request. Students who want stronger answers can choose a paid model, from the least to the most expensive, through a Pro subscription that covers the cost of those models.
Options	Option 1: Do nothing. Students keep combining existing tools: a PDF reader, a general AI assistant or NotebookLM for explanations, and Anki or Quizlet for practice. Option 2: Build the AI Learning Environment as a desktop app with free models and a model picker. An Electron + React app reads the student's local course folders and opens PDFs in a built-in reader. Only the passage the student selects is sent to an AI model; original files stay on the device. • Analysis of the passage, flashcards and quizzes run on free cloud AI models (for example the Gemini API free tier, Groq or OpenRouter's free models). • For explanations, the student picks a model from a list sorted from least to most expensive. The free model is the default; paid models (for example Claude Haiku 4.5, Claude Sonnet 5.5 and Claude Opus 5.5) are unlocked by a Pro subscription. • Fall 2026 (ENSE 400): PDF reader, AI explanation with page reference and model picker, flashcards and quizzes. • Winter 2027 (ENSE 477): accounts and the Pro subscription, spaced review against exam dates, voice input, iPad version. Option 3: Build the same app on one paid model for everyone. Every explanation, flashcard and quiz uses a single paid model (for example Claude Sonnet 5.5), with the team paying all AI costs. Option 4: Build the same app with a fully on-device AI model. Explanations and practice come from an open-weight model running on the student's laptop (for example through Ollama or llama.cpp), so no text leaves the device.
Cost-Benefit Analysis	
Option 1: Do nothing Costs • The student problem stays: context is re-supplied every session, answers are not tied to a page, and practice is made by hand in a separate app. • The research done so far (student interview, competitor analysis, user analysis) goes unused, and the team needs a new capstone project. Benefits • No development time, financial cost or technical risk. • Students already know the existing tools, and several have free tiers. Option 2: Free models with a model picker (Pro subscription for paid models) Costs • Time: two students at roughly 8 to 10 hours a week each over two terms, about 400 to 500 person-hours in total.	

This option adds work for connecting several model providers, user accounts, subscription payments and per-user usage limits.

- **Money (self-funded):** development, testing and the free tier cost nothing per request. Testing the paid models is estimated at about US\$50 across both terms. At current Claude prices, one explanation costs about half a cent on Haiku 4.5 (US\$1 input / US\$5 output per million tokens), about 1 cent on Sonnet 5.5 (US\$2 / US\$10) and about 2.5 cents on Opus 5.5 (US\$4 / US\$20). Accounts, the server and the database can run on free hosting tiers; the payment processor takes a fee of about 3% per payment. The Apple Developer Program (US\$99 a year) is needed only if the iPad version goes ahead.
- **Risks:** free cloud tiers have rate limits that are shared by every student using the team's key, and their terms can change; some free tiers allow the provider to use prompts to improve their models, which must be disclosed to students; free models may give weaker answers; Pro pricing needs monthly usage limits so the most expensive models cannot cost more than a subscription brings in; payments add security work; the scope is large for a two-person team.

Benefits

- Free for students to start, so cost is no barrier compared with free general assistants.
- Students choose their own trade-off between cost and answer quality, with prices shown in the picker.
- The team pays nothing for free-tier requests, and paid-model costs are covered by subscription revenue, which suits a bootstrapped project.
- One model-independent AI layer means new models, including on-device ones, can be added later without a rewrite.
- Removes the repeated setup step, gives every explanation a page reference, and generates practice from the same passage.
- Original files stay on the device; only the selected passage is sent, and course files are never stored on a server.

Option 3: One paid model for everyone

Costs

- The team pays for every request, including those from students who never pay. For example, 100 active students asking 300 questions a month on Claude Sonnet 5.5 would cost roughly US\$300 to US\$400 a month, which a self-funded team cannot sustain.
- Students cannot pick a cheaper or stronger model.

Benefits

- Consistent answer quality from one well-tested model.
- Simplest to build: one provider, and no accounts or payments needed for the course demo.

Option 4: Fully on-device AI

Costs

- Same development time as Option 2, plus extra time to package, test and tune a local model across different student laptops.
- Answers are expected to be slower and less accurate on typical student laptops (still to be confirmed by testing), and the model needs several gigabytes of disk space and memory.
- An iPad version becomes much harder.
- Highest technical risk: local AI performance is the project's main unknown, and a failure here would put the fall demo at risk.

Benefits

- No text ever leaves the device, which is the strongest privacy position.
- No per-request AI cost, and it works offline.

Recommendation

Option 2: build the AI Learning Environment as an Electron + React desktop app that runs on free cloud AI models by default and lets students pick a paid model for explanations, from the least to the most expensive, through a Pro

subscription.

It removes the cost barrier for students, keeps the team's AI costs near zero while self-funded, and lets the paid tier pay for itself. Option 3 is too expensive for a bootstrapped team, and Option 4 carries too much technical risk for the fall deliverables.

The work is phased to protect the fall demo. In ENSE 400 the model picker is built with free models and a few paid models the team pays for during testing. Accounts, the Pro subscription and payments follow in ENSE 477. Because the AI layer is model-independent, an on-device model (Option 4) can still be tested and added later.
